VERIFICATION · APPENDIX TO JOB SIGNAL ACQUISITION

Crawlability Audit

Sixteen sources checked against their own published rules rather than assumption. The tier structure held; three details did not, and one of them adds a source that had been written off.

Checked 10 Sep 2026 **Sources** 16 **Method** live robots.txt + endpoint fetch

§1 Method

Each source's `robots.txt` was fetched live and read for four things: rules affecting job, career and employment paths; `Crawl-delay` directives; blanket `Disallow: /` by user-agent; and published sitemaps. Two API endpoints were called directly to confirm they respond without authentication.

This audit reports **stated policy and observed response**. It is not legal advice, and a permissive `robots.txt` is not a licence — a site's terms of use are a separate document and should be read for any source that becomes load-bearing.

7	5	3	1
OPEN	CONDITIONAL	PROHIBITED	UNREACHABLE

§2 Results

OPEN — NO POLICY OBSTACLE TO THE JOB PATHS

SOURCE	STATUS	WHAT THE FILE SAYS	PLATFORM
State Bar of California calbar.ca.gov	OPEN	No job/career path restrictions. Selective `Disallow` on admin paths only, with `Allow` rules for assets. No crawl-delay.	Drupal /core/, /profiles/ paths
New York State Bar nysba.org	OPEN	Only `/wp-admin/` disallowed. Two sitemaps published. No crawl-delay.	WordPress
NALA nala.org — paralegal association	OPEN	Only `/wp-admin/` disallowed. Sitemap published. No crawl-delay.	WordPress
Chicago Bar Association chicagobar.org	OPEN	No job path restrictions. Blocks only commercial SEO crawlers — Ahrefs, Semrush and variants, MJ12, Yandex. No crawl-delay.	iMIS /AsiCommon/, /iParts/
LA County Bar lacba.org	OPEN with a real constraint	No job path restrictions. Sitemap published. **`Crawl-delay: 5`** applies to general and allowlisted agents. Large explicit bad-bot blocklist.	—
Craigslist craigslist.org	OPEN more than expected	No blanket disallow. **No restriction on `/search/` or job-category paths.** Only interaction endpoints blocked — `/reply`, `/flag`, `/suggest`. 30 sitemaps published.	—
Greenhouse boards API boards-api.greenhouse.io	VERIFIED LIVE	Returns JSON without authentication. Per job: `id`, `title`, `location`, `absolute_url`, `requisition_id`, **`first_published`**, **`updated_at`**, `application_deadline`.	Public ATS API

> THE GREENHOUSE RESPONSE SETTLES A DESIGN QUESTION
>
> It returns both `first_published` and `updated_at` as separate fields. That is exactly the pair repost detection needs — a listing whose `updated_at` moves while `first_published` stays fixed is a refreshed vacancy, and here that is readable directly rather than inferred by holding our own history.

CONDITIONAL — REACHABLE, BUT WITH A STATED LIMIT OR AN ACCESS PROBLEM

SOURCE	STATUS	FINDING
The Florida Bar floridabar.org	CONDITIONAL	`Allow: /` for `User-agent: *` and no job-path restrictions — but explicit `Disallow: /` for **ClaudeBot, GPTBot, CCBot, Google-Extended, Amazonbot, Bytespider, Applebot-Extended, meta-externalagent**, plus the header `Content-Signal: search=yes, ai-train=no, use=reference`. They have drawn a line at AI training, not at crawling. A conventional, honestly-identified crawler is not in the blocked set — but the posture is deliberate and should be respected as stated.
Lever postings API api.lever.co	PARTIAL	Endpoint responded without authentication, but the test slug returned an empty array, so the per-posting field structure could not be confirmed in this check. Re-test against a slug with live postings before relying on specific fields.
NFPA paralegals.org	BLOCKED AT EDGE	HTTP 403 on `robots.txt` itself. No policy could be read — this is infrastructure refusing the request, not a published rule. See §4.

SOURCE	STATUS	FINDING
American Bar Association americanbar.org	BLOCKED AT EDGE	HTTP 403 on `robots.txt`. Same situation as NFPA.
State Bar of Texas texasbar.com	UNREACHABLE	Connection refused at the network layer from this egress. **No conclusion should be drawn** — this is not evidence of a policy either way. Re-check from the crawler's actual network before planning around it.

§3 Closed sources — in their own words

The three large boards are not ambiguous, and this is no longer a judgement call. Each publishes a rule that covers exactly the paths the pipeline would need.

```
INDEED  indeed.com/robots.txt · User-agent: *
  Disallow: /job/
  Disallow: /jobs/          -- plus country variants /jobs/US/ etc.
  Disallow: /viewjob?
  -- no blanket disallow, but the three disallowed prefixes are
  -- precisely the listing and detail paths. nothing usable remains.

ZIPRECRUITER  ziprecruiter.com/robots.txt · User-agent: *
  Disallow: /
  Allow: /llms.txt
  -- deny by default. the single permitted file is a manifest.

LINKEDIN  linkedin.com/robots.txt · header text
  "The use of robots or other automated means to access LinkedIn
  without the express permission of LinkedIn is strictly
  prohibited."
  -- no User-agent: * section at all. named agents are disallowed
  -- from /jobs?runSearch*, /jobs-guest/, /api/jobPostings/jobs*,
  -- /jsearch*. access is by whitelist application only.
```

THIS CONFIRMS THE ARCHITECTURE RATHER THAN CONSTRAINING IT

The build order in the acquisition spec already put these last, on the reasoning that small firms mostly avoid paid boards. That reasoning is now joined by a harder one: for these three, **a licensed data provider is not the compliant alternative to scraping, it is the only route that exists.**

The design consequence is unchanged. Lane 1 is the foundation; lane 2 is a purchase, made against a measured coverage gap.

§4 Stated policy is not the same as an edge block

Three sources in this audit failed for reasons that look identical from the client side and are completely different in meaning. The distinction has to be carried in the source registry, or the crawler will treat a temporary network condition as a permanent prohibition.

OBSERVATION	MEANS	CORRECT RESPONSE
robots.txt read, job paths disallowed Indeed, ZipRecruiter, LinkedIn	A published rule. Stable, deliberate, and the site's stated position.	Do not crawl. Mark `policy_closed`. Route to a licensed provider or drop the source.
HTTP 403 on robots.txt itself NFPA, ABA	A WAF or bot-management layer refusing this request — commonly a datacentre IP range. **No policy was published or read.**	Mark `edge_blocked`, not closed. Re-attempt from the crawler's real egress with an honest identifying user-agent. If it still fails, deprioritise — do not attempt to defeat it.
Connection refused Texas Bar	Network-layer failure. Could be geography, routing, an outage, or IP reputation.	Mark `unreachable` and retry on a schedule. Draw no conclusion about policy.

> **DO NOT COLLAPSE THESE THREE INTO ONE FLAG**
>
> A single `blocked` boolean loses the distinction that matters: one is a decision to respect permanently, the others are conditions to retry. Conflated, the crawler either abandons reachable sources or keeps hammering ones that have said no.

§5 What changed against the assumptions

ASSUMPTION IN THE ACQUISITION SPEC	AUDIT FINDING	CHANGE
Bar and paralegal association boards are primary sources	Confirmed. Four of five reachable association sites carry no job-path restriction; two publish sitemaps.	HOLDS
Craigslist is a light-touch supplementary feed	**Understated.** No blanket disallow, job-category and search paths unrestricted, 30 sitemaps published. It is more open than most association sites.	PROMOTE TO LANE 1
Parsers per platform, not per board, gives roughly 3:1 reuse	Supported. Three platforms already appear across five sites — WordPress twice, iMIS once, Drupal once.	HOLDS
Public ATS endpoints return structured job data	Confirmed for Greenhouse, with better fields than expected. Lever partially confirmed only.	HOLDS Re-test Lever
The big three are "buy, don't scrape"	Confirmed explicitly by all three, in their own published files.	HOLDS
Crawl rate of one request per two seconds is adequate	**Not universal.** LA County Bar declares `Crawl-delay: 5`.	PER-HOST, HONOUR DECLARED
Association sites are uniformly reachable	**Not so.** Two returned 403 at the edge and one refused connection — three of eleven attempted.	ADD REACHABILITY STATES

§6 Design impact

Five changes to the acquisition components, all small and all cheaper to make now than after the crawler is written.

1. SOURCE REGISTRY gains a policy state, not a boolean

```
source_state ∈ { open, rate_limited, policy_closed,
                 edge_blocked, unreachable, auth_required }
crawl_delay_s   int      -- parsed from robots.txt, per host
robots_checked  date     -- staleness of our own reading
platform        text     -- wordpress | imis | drupal | naylor …
```

2. CRAWL-DELAY IS READ, NOT ASSUMED

```
the per-host token bucket takes its rate from the host's own
declared Crawl-delay where one exists, and falls back to the
2s default only where none is declared.
-- lacba at 5s: a 200-listing board takes ~17 min. across 60
-- metro bars that is real scheduler time and has to be planned,
-- not discovered on the first slow night.
```

3. USER-AGENT IS HONEST AND CONTACTABLE

```
one identifying UA string with a contact URL. never rotated to
evade, never impersonating a browser or a named search bot.
-- florida bar blocks AI-training agents by name while allowing
-- general crawling. that distinction only works if crawlers
-- identify honestly — and it is the direction more sites are
-- moving. Content-Signal will appear on more of these.
```

4. CRAIGSLIST MOVES INTO LANE 1

```
per-city RSS by job category, metros first.
-- expect high agency noise here — the §4 agency filter matters
-- more on craigslist than on any association board.
```

5. GREENHOUSE first_published IS USED DIRECTLY

```
where an ATS supplies both first_published and updated_at,
read repost signal from the source instead of inferring it.
our own first_seen is still recorded — for every other source,
and as a cross-check on this one.
```

§7 Still unchecked

This audit covered sixteen sources. The registry needs roughly a hundred and ten. What remains, in priority order:

GROUP	COUNT	PRIORITY	NOTE
Remaining state bars	~46	HIGH	Expect a handful of platforms to cover most. Identify the platform per site while checking — it determines parser reuse.
Metro & county bars	~55	HIGH	Top 20 metros first. Closest to the ICP of any source.
State paralegal associations	~30	MEDIUM	Role-specific and low noise, but small volume each.
Remaining ATS platforms	4	MEDIUM	Workable, JazzHR, BambooHR, Recruitee — confirm public endpoint shape for each.
Re-check the three that failed	3	HIGH	Texas Bar, NFPA, ABA — from the crawler's real egress, not this one.

> MAKE THE AUDIT A COMPONENT, NOT A TASK
>
> Checking a hundred and ten sites by hand is a week that produces a snapshot which starts decaying immediately. The same work as a script — fetch `robots.txt`, parse the directives, fingerprint the platform, write to the source registry — runs in an afternoon and can be re-run on a schedule.
>
> It also becomes the mechanism for §8. A hand-built table is a document; a generated one is infrastructure.

§8 Re-audit

These files change. The Florida Bar's AI-crawler blocklist and its `Content-Signal` header did not exist a few years ago, and that pattern is spreading. A permission checked once and never rechecked becomes an assumption again.

TRIGGER	ACTION
Monthly, all registered sources	Re-fetch `robots.txt`, diff against the stored copy, alert on any change touching job paths, crawl-delay, or agent-level rules.
A source's yield drops to zero	Check `robots.txt` before assuming the parser broke. A newly added `Disallow` and a site redesign present identically.
Repeated 403 or 429 from a source	Stop, do not retry harder. Re-read the policy, lower the rate, and if it persists mark the source closed rather than working around it.
Before a source becomes load-bearing	Read the site's terms of use, not only `robots.txt`. The two are separate documents and the terms are the binding one.

Checked 10 September 2026 by live fetch of each source's robots.txt, plus direct unauthenticated calls to two ATS endpoints. Findings reflect that moment only.

This records stated policy and observed response. It is not legal advice. A permissive robots.txt is not a licence — terms of use are a separate document and govern independently. Three sources could not be read: two returned HTTP 403 at the edge and one refused connection, and none of the three should be treated as having stated a policy.